



SCHOOL OF COMPUTING
FACULTY OF ENGINEERING
UNIVERSITI TEKNOLOGI MALAYSIA

FINAL YEAR PROJECT (PSM) STUDENT LOG BOOK

PROJECT TITLE

DESIGN AND DEPLOYMENT OF A SECURE CLOUD-BASED
PATIENT DATA MANAGEMENT SYSTEM USING A THREE-TIER
ARCHITECTURE ON AWS

STUDENT INFO

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Department : IS / SE / **CS**

SUPERVISOR INFO

Main Supervisor : Johan Mohamed Sharif

Email : johan@utm

Co-Supervisor (if applicable):.....

STUDENT LOG BOOK NOTES

- 1) This log book need to be used by a PSM 1 and PSM 2 student for a purpose of reporting all progress of their PSM projects.
- 2) It is the responsibility of the student that this log book is **kept up to date** and that the student complies with the Supervisor's suggestions and recommendations as noted by the student in the log book and approved by the Supervisor.
- 3) Student **MUST** makes a regular meeting with a supervisor at least **ONCE** in **TWO WEEKS**. A minimum number of meetings for PSM student-supervisor meeting are 6 times per semester.
- 4) A PSM Department Coordinator and supervisor have a right for not to allow the student to present their project if this regulation is not complied.
- 5) This log book has to be submitted together with the proposal report (for PSM 1) and the final thesis report (for PSM 2) as stated in the PSM Activities Calendar.
- 6) The Faculty of Computing reserves right not to accept thesis for examination if this log book is not properly documented.



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PSM LOG BOOK

SEMESTER: 20252026-2 TYPE: **PSM 1** / PSM 2

Date :26/03/2026 Meeting: **1** / 2 / 3 / 4 / 5/ 6 / 7

Student
(Meeting Minute/
Achievements/
Activities)

: - **Title:** Discussed PSM1 Proposal title and system development
 - **Stakeholder Analysis:** Defined the primary stakeholder, Alamin Clinic (Saudi Arabia), and used their recent ransomware attack as a core case study for system design
 - **Proposed Architecture:** Finalized and Searched the design of a Three-Tier Clinic Record Management System on AWS VPC

Supervisor
(Suggestion&
Comments)

: - **Scope Management:** Advised focusing on core functional modules
 - **Validation Metrics:** Encouraged putting a metrics to validate the system
 - **Objectives:** Advised on the objective academic writing

Next Meeting Plan

: Review PSM1 proposal Changes

Supervisor's Signature

..... Date:



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PSM LOG BOOK

SEMESTER: 20252026-2 TYPE: **PSM 1** / PSM 2

Date : 02/04/2026 Meeting: 1 / **2** / 3 / 4 / 5 / 6 / 7

Student : - **Edited** Objectives based on comments
(Meeting Minute/ - **Specified** and narrowed the scope
Achievements/ - **Added** more on Problem statements and proposed solution
Activities)

Supervisor : - **Recommended** to add Literal review on existing works
(Suggestion& - **Advised** on referencing with no more than 5 to 10 years
Comments) - **Advised** on comparing existing and proposed system

Next Meeting : Review Edits based on comments and Processed with Progress 1 meeting
Plan

Supervisor's
Signature Date:



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PSM LOG BOOK

SEMESTER: 20252026-2 **TYPE:** **PSM 1** / PSM 2

Date : 10/04/2026 **Meeting:** 1 / 2 / **3** / 4 / 5 / 6 / 7

Student : - **Meted** to briefly review the final proposal outcome
(Meeting Minute/ - **Presented** and no farther comments given
Achievements/
Activities)
Supervisor : N\A
(Suggestion&
Comments)

Next Meeting : Progress 1
Plan

Supervisor's
Signature **Date:**



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PSM LOG BOOK

SEMESTER: 20252026 **TYPE:** **PSM 1** / PSM 2

Date : 20/04/2026 **Meeting:** 1 / 2 / 3 / **4** / 5 / 6 / 7

Student
**(Meeting Minute/
Achievements/
Activities)**

: - **Case Study Identification:** Defined Alamin Clinic (Saudi Arabia) as the primary stakeholder after analyzing their specific ransomware incident

- **Problem Background:** Drafted the initial problem statement focusing on the flaws of manual server configuration and the lack of network segmentation

- **Literature Initiation:** Commenced research into the state of cloud computing in healthcare, specifically looking at studies from the early 2010s to present

Supervisor
**(Suggestion&
Comments)**

: - **Gap Analysis:** Encouraged me to focus on the "market gap" for small healthcare providers who cannot afford enterprise solutions like Epic Systems

- **Security Standards:** Advised using the HIPAA framework as a "gold-standard" benchmark for evaluating the proposed system.

- **Organizational Context:** Suggested mapping out the clinic's hierarchy to better design the future Role-Based Access Control (RBAC)

Next Meeting Plan

: **Analysis Framework:** Apply the 5W1H framework to decompose the clinic's security vulnerabilities

Complete the technology review focusing on Terraform (IaC) and DevSecOps.

Supervisor's Signature

..... **Date:**



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PSM LOG BOOK

SEMESTER: 20252026 TYPE: **PSM 1** / PSM 2

Date : 30/04/2026 Meeting: 1 / 2 / 3 / 4 / **5** / 6 / 7

Student (Meeting Minute/ Achievements/ Activities) : - **Chapter 1 Finalization:** Finalized the Project Aim, Objectives, and Scope, ensuring a clear distinction between in-scope (VPC, IAM, IaC) and out-of-scope (Billing, Pharmacy) modules
 - **Problem:** Completed the 5W1H analysis of Alamin Clinic's vulnerabilities
 - **Literature Completion:** Finalized the technology review section, linking "Shift-left" security principles to the proposed DevSecOps pipeline
 - **Comparison Table:** Constructed a comparative analysis between traditional on-premise systems, OpenEMR, and the proposed cloud solution

Supervisor (Suggestion& Comments) : - **Submission Ready:** Approved the final draft of Chapters 1 and 2 for the Progress 1 evaluation

Next Meeting Plan : **Methodology Focus:** Begin drafting Chapter 3 (System Development Methodology) with a focus on the Agile phases and the Terraform implementation plan
Progress 1 Review resubmission if needed

Supervisor's Signature Date:



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PSM LOG BOOK

SEMESTER: 20252026 **TYPE:** **PSM 1** / PSM 2

Date :15/05/2026 **Meeting:** 1 / 2 / 3 / 4 / 5/ **6** / 7

**Student
(Meeting Minute/
Achievements/
Activities)** : Presented the methodology chapter draft to supervisor. Completed the justification for selecting Agile with DevSecOps over Waterfall and Scrum methodologies, supported with comparison table. Defined the five-sprint structure with security gates for each sprint. Completed the technology stack description covering AWS services, Terraform, Docker, GitHub Actions, Trivy, SonarQube, and Checkov. Completed functional requirements (FR-01 to FR-12) and non-functional requirements (NFR-01 to NFR-11) with verification methods. Completed minimum client-side and server-side requirements tables.

**Supervisor
(Suggestion&
Comments)** : Methodology justification is well-structured and clearly argued. Suggested ensuring that each sprint's security gate is measurable and tied to a specific tool output. Recommended adding a requirements traceability diagram to link functional requirements back to the identified system vulnerabilities from Chapter 2. Noted that the technology descriptions should reference relevant literature where possible.

**Next Meeting
Plan** : Present Chapter 4 draft covering system architecture design, VPC network topology, and security group configuration. Begin use case analysis and user story documentation.

**Supervisor's
Signature** **Date:**



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PSM LOG BOOK

SEMESTER: 20252026 **TYPE:** **PSM 1** / PSM 2

Date :22/05/2026 **Meeting:** 1 / 2 / 3 / 4 / 5/ 6 / **7**

Student

**(Meeting Minute/
Achievements/
Activities)**

Presented the completed Chapter 4 draft to supervisor. Completed the full system architecture design including three-tier AWS VPC with six subnets across two Availability Zones. Completed security group rules, NACL configuration, and IAM role definitions with least-privilege policies. Completed the DevSecOps pipeline design with six stages and blocking gates for Trivy, SonarQube, and Checkov. Completed the database schema with six tables and PostgreSQL row-level security policies for Doctor, Patient, and Admin roles. Completed the HIPAA Security Rule compliance mapping table. Completed interface wireframes for Login Screen, Doctor Dashboard, Admin Dashboard, and Patient Portal.

Supervisor

**(Suggestion&
Comments)**

: Overall design is comprehensive and well-documented. Suggested clarifying the distinction between Security Groups and NACLs in the text to highlight their complementary roles. Recommended ensuring the RLS policies are explicitly tied back to the HIPAA access control requirement in the compliance mapping table. Noted that the interface wireframes clearly demonstrate role segregation, which strengthens the system's security-by-design argument.

**Next Meeting
Plan**

: discuss any corrections

**Supervisor's
Signature**

..... Date:

